

Text Vectorization & The Polysemy Barrier

Core Objective

To programmatically demonstrate the foundational limits of **Bag-of-Words / Static Count Vectors** when processing human language. You will use pandas and scikit-learn to filter a real-world Kaggle dataset and prove mathematically why static representations fail to capture **context** when a word changes meaning (polysemy). This directly establishes the technical necessity for the **Self-Attention Mechanisms** used in modern Generative AI models.

Assigned Dataset

- **Dataset :** <https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews>
- **File to use :** Reviews.csv

Step-by-Step Lab Workflow

Step 1: GitHub & Environment Setup

- Create a public GitHub repository named **[Name]-gen ai-foundations**.
- Clone the repository to your local machine.
- Download Reviews.csv from the Kaggle link and place it inside your project folder.
- Create a Jupyter Notebook named nlp_vector_lab.ipynb and a markdown file named answers.md.

Step 2: The Coding Challenge

Open nlp_vector_lab.ipynb. You must write a Python script from scratch that implements the following technical specifications. Do not hardcode vectors; use library methods.

Technical Specifications:

1. **Data Ingestion:** Load the first 50,000 rows of Reviews.csv using pandas.
2. **Text Selection:** Create a list containing exactly three specific text strings where the word "light" is used in different contexts:
 - *Sentence 1:* A review where "light" means low-calorie, weight, or texture (e.g., "This olive oil is wonderful and very light on the stomach.").
 - *Sentence 2:* A review where "light" means illumination or physical visibility (e.g., "The packaging was broken and let too much light inside the box.").
 - *Sentence 3:* Another review where "light" matches the low-calorie or weight context of Sentence 1 (e.g., "I prefer this brand because it is a light and healthy snack.").
3. **Vectorization Engine:** Import and initialize a standard word-counting text vectorizer from scikit-learn. Transform your list of 3 sentences into a document-term matrix, and explicitly convert that matrix into a standard dense NumPy array.
4. **Vector Slicing:** Extract the individual row vectors for each of the three sentences from your dense matrix. Assign them to distinct variables: vec_1, vec_2, and vec_3.
5. **Similarity Mathematics:** Calculate the mathematical similarity between **Sentence 1** and **Sentence 2** (Different Contexts).

- Calculate the mathematical similarity between **Sentence 1** and **Sentence 3** (Similar Contexts).
 - **Constraint:** You must use the linear algebra **Dot Product** function from the **numpy library** to compute these scores.
6. **Console Output:** Print both calculated similarity scores clearly to the console.

Step 3: Theory & Reflection Questions

Open answers.md and provide a 3-to-4 sentence analytical response for each question:

Question 1: The Vector Conflict

Review your printed dot product similarity scores. Did your code show a significantly high similarity score between the two contextually different uses of the word "light"? Explain why a basic frequency-counting vectorizer calculates a false mathematical relationship here.

Question 2: The Context Blindspot

From a Data Science perspective, explain why this bag-of-words approach creates a massive bottleneck for building search engines or chatbots. What linguistic property is completely lost when text is turned into static counts?

Question 3: The GenAI Architectural Fix

How do modern Large Language Models (like GPT or LLaMA) solve this exact problem? Mention how **Masked Self-Attention** and **Context-Aware Embeddings** work together to ensure the word "light" gets completely different mathematical vectors in those three sentences.

Step 4: Git Submission Workflow

Open your terminal inside your project directory and run these commands to deploy your work to GitHub:

Bash

git init

git remote add origin <YOUR_GITHUB_REPOSITORY_URL>

git add .

git commit -m "MCA Lab: Completed Amazon Review vector analysis and reflection"

git branch -M main

git push -u origin main

Submit your Github link to the excel file shared with it.